

Risk Analysis and Management

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Chapter 6 • Software Engineering

What is Risk?

A risk is a potential problem — it MIGHT happen, or it might NOT.

Uncertainty

A risk may or may not occur.
Nothing is 100% certain.
(If it's certain, it's a constraint — not a risk!)

Loss

If a risk becomes real, something bad happens:
costs increase, schedules slip,
or the project fails.

Types of Risk



Project Risk

Threats to the plan — if these happen, the schedule slips and costs go up.
Example: key team member quits.



Technical Risk

Threats to quality and delivery of software — implementation may become very hard.
Example: unproven technology chosen.



Business Risk

Threats to whether the product is even worth building.
Example: nobody wants the product (market risk).

Reactive vs Proactive Strategies

REACTIVE

"Don't worry, I'll think of something!"

- Do nothing until problems hit
- Then panic and start firefighting
- Crisis management is the norm
- Most teams unfortunately use this!

VS

PROACTIVE

Plan ahead. Expect problems before they hit.

- Identify risks early
- Analyze each risk's probability & impact
- Have a backup plan (contingency) ready
- Control risks before they become disasters

4 Steps to Manage Risk

1

Identify

List everything that could go wrong. Ask: What might hurt this project?

3

Rank

Sort risks by their probability $\times$ impact. Focus on the most dangerous ones first.

2

Analyze

For each risk, estimate: How likely is it? How bad would it be if it happens?

4

Plan

Create a contingency plan for high-risk items. Know what to do if it goes wrong.

4 Types of Risk Impact

Performance Risk

Will the software actually do what it's supposed to do? Meeting requirements and being usable.

Cost Risk

Will the project stay within budget? Cost overruns are very common in software projects.

Support Risk

Will the software be easy to fix and update later? Poor code means expensive maintenance.

Schedule Risk

Will the project be delivered on time? Delays affect business, revenue, and customer trust.

The RMMM Strategy

Risk Mitigation · Monitoring · Management

Mitigation (Avoid)

Try to PREVENT the risk from happening in the first place.

Example: Worried about staff leaving? Cross-train people, document everything, assign backup staff for every key role.

Monitoring (Watch)

Keep an eye on warning signs. Are there indicators that a risk is getting worse?

Example: Track team morale, look for signs someone might quit soon.

Management (React)

If mitigation failed and the risk happened — execute your contingency plan!

Example: Someone quit? Activate the backup person immediately.

7 Principles of Risk Management



See the big picture — risks exist within a bigger business context.



Keep updating — risks change; add new ones as you learn more.



Look ahead — think about future risks and make contingency plans.



Share the vision — all stakeholders need a common goal to spot risks better.



Talk openly — everyone should be able to raise a risk without fear.



Work as a team — combine everyone's skills to manage risks together.



Integrate risk thinking into every step of the software process.

Thank You

